TEXORA · LECTURE SHEET

Present Trends of Man-Made Fibre Production

Where the industry stands and where it is moving: polyester's dominance, Asia's production map, the recycled and bio-based shifts, microfibre pressure, and what the trends mean for a cotton-and-jute country like Bangladesh. (All figures in this sheet are deliberately approximate — orders of magnitude from recent industry reporting, not audited statistics.)

TE-2108 · MAN-MADE TEXTILE FIBERS · TOPIC 5 · LEVEL 2 TERM 1

1. The global picture

World fibre production is of the order of **120 million tonnes a year** in the early 2020s and still growing at a few percent annually. Man-made fibres supply **roughly seven-tenths** of that total — the crossing point where synthetics passed all natural fibres combined came decades ago, and the gap keeps widening because man-made output can expand with factory capacity while cotton is fenced in by land and water.

- **Polyester alone is roughly 55 % of all fibre** — more than cotton, wool, and all other synthetics each by a wide margin; its staple and filament plants are the industry's centre of gravity.
- **Cotton holds roughly a fifth** of world fibre output, its share slowly eroding even when absolute tonnage grows.
- **Man-made cellulosics** (viscose, lyocell, modal, acetate) are roughly 5–6 % — small in tonnes, strategically important as the renewable-polymer bridge.
- Nylon, acrylic, polypropylene and elastane divide most of the remainder; wool is only about 1 % of world fibre by weight despite its price and prestige.

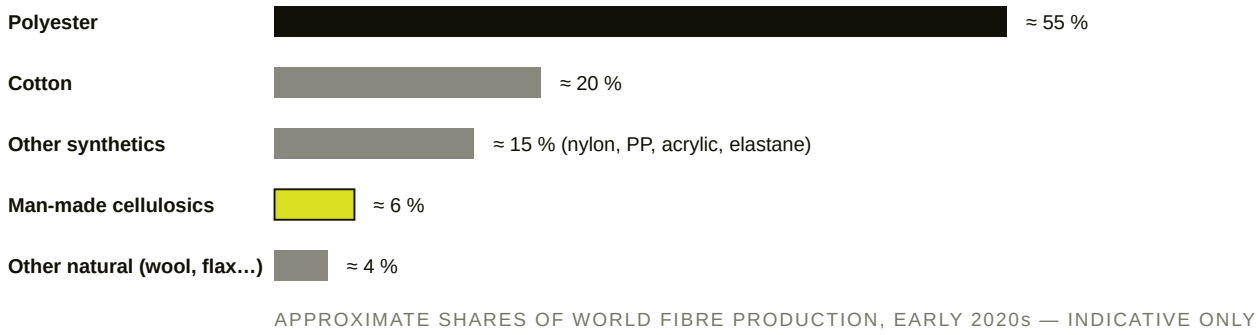


Figure 1 — The world's fibre diet in round numbers. One polymer — polyester — outweighs every natural fibre put together; man-made fibres of all kinds supply roughly 70 %.

2. The production map

Production has concentrated in Asia for a generation. **China produces well over half of the world's man-made fibre** — by many estimates around two-thirds of polyester — with India, Southeast Asia (Vietnam, Indonesia), Taiwan and South Korea holding most of the remainder; the old centres in the USA, Europe and Japan now specialise in high-performance and speciality grades (Topic 4) rather than commodity tonnage. The trend inside Asia continues toward **mega-plants integrated from PTA and MEG through polymerisation to spinning and texturing on one site**, which squeezes independent small producers everywhere — including the import bill of countries that buy their synthetic yarn and fabric from these complexes.

3. The sustainability turn

3.1 Recycled fibres — growing, but from a small base

Recycled fibres are still **under a tenth** of world fibre supply, and most of that is one stream: **recycled PET made from bottles**, not from old garments. Bottle flake is clean and available; textile waste is blended, dyed and contaminated, so textile-to-textile recycling (mechanical and chemical, TE-2107 Topic 6) is only now scaling. Expect regulation — extended producer responsibility and recycled-content requirements in the EU and elsewhere — to pull this share upward through the decade, and expect brands to compete for the limited bottle-grade feedstock with the bottle industry itself.

3.2 Cellulosics with better chemistry

Within regenerated fibres the growth is in **lyocell and branded closed-loop viscose**: pulp from certified forests, solvent recovery above 99 %, and audited supply chains. Commodity viscose capacity keeps growing too, but new plants are expected to run CS_2 recovery as standard — the "dirty viscose" era is closing under buyer pressure.

3.3 Bio-based and next-generation fibres

Bio-based routes (bio-MEG for partly plant-based PET, PLA, bio-based nylons from castor oil) hold a small but visible share; fermentation- and algae-based experimental fibres attract investment far beyond their tonnage. The honest trend line: these are options being bought for the future, not yet volume materials.

3.4 Process trends

- **Dope (solution) dyeing** expands steadily — pigmenting the melt eliminates the dyehouse for those shades, saving nearly all dyeing water and giving the highest fastness; black, navy and workwear shades lead.
- **Microfibres and functional variants** (flame-retardant, antimicrobial, high-modulus low-shrinkage industrial yarns) grow faster than commodity tonnage — value migrates from tonnes to performance.
- **Nonwovens** (spunbond/meltblown polypropylene and polyester) are among the fastest-growing outlets for man-made fibre, driven by hygiene, medical and filtration demand.
- **Microfibre-shedding pressure** (TE-2107 Topic 6) is becoming a specification issue: low-shed constructions, filament rather than staple formats, and test standards are entering buyer requirements.

> **WORKED EXAMPLE — READING GROWTH RATES**
>
> Industry output grows at roughly 3 % a year. Doubling time = ln 2 ÷ ln 1.03 = 0.693 ÷ 0.0296 ≈ **23 years**.
>
> So at trend growth, a further ≈ 120-million-tonne industry becomes ≈ 240 million tonnes within one professional generation — which is why every percentage point of recycled content and every closed-loop plant matters at a scale small pilot projects cannot show. Conversely, a fibre holding a constant tonnage in a growing market is *losing* share: cotton's roughly steady output against a growing total explains its sliding percentage in Figure 1 without any farmer harvesting less.

4. What the trends mean for Bangladesh

Bangladesh's apparel industry — the world's second-largest garment exporter — was built on imported cotton and, increasingly, imported man-made inputs: domestic spinning covers cotton yarn well, but **most synthetic filament, much synthetic staple and most MMF fabrics are still imported**, mainly from China and India. The trend consequences are direct:

- Global demand is shifting toward MMF garments (roughly 70 % of world apparel fibre demand is man-made), so an export industry concentrated in cotton captures a shrinking slice of the growing

market — diversifying into MMF apparel is a stated national industrial priority.

- Local polyester and filament projects have begun to narrow the gap, but scale, power costs and feedstock (imported PTA/MEG or chips) still decide competitiveness against the integrated Asian mega-plants of §2.
- Buyer sustainability rules (recycled content, closed-loop cellulosics, chemical management) arrive in Dhaka as order conditions, not as theory — compliance with them is becoming a condition of market access for Bangladeshi factories.

Glossary

Extended producer responsibility (EPR) — rules making brands/manufacturers finance the collection and recycling of what they sell. **Feedstock** — the raw input stream (here: bottles for recycled PET, pulp for cellulosics). **Integrated plant** — a site running the whole chain from petrochemical intermediates to spun yarn. **PTA / MEG** — purified terephthalic acid and mono-ethylene glycol, the polyester monomers. **Spunbond / meltblown** — nonwoven processes laying filaments directly into a web.

Key points

• World fibre output ≈ 120 Mt and growing; man-made ≈ 70 % of it, polyester alone ≈ 55 %; cotton's share slides even when its tonnage does not.
• Production is concentrated in Asia, above all China, in ever-larger integrated plants; the West keeps the speciality end.
• Recycled content is still under 10 % and mostly bottle-PET; textile-to-textile recycling and closed-loop cellulosics are the growth frontier, pushed by regulation.
• Value migrates to performance: dope-dyed, functional, microfibre and nonwoven outlets grow faster than commodity tonnage.
• For Bangladesh, the MMF shift is an industrial-strategy question: import dependence today, diversification and compliance as the conditions of tomorrow's orders.

Practice

1. Using the doubling-time method of the worked example, estimate the doubling time for a segment growing at 6 % a year (functional fibres). ln 2 ≈ 0.693, ln 1.06 ≈ 0.0583.
2. A brand demands 50 % recycled content in its polyester garments. Explain why meeting this from bottle-PET alone becomes harder as more brands make the same demand, naming the competing claimant for the feedstock.
3. Give two reasons why dope dyeing grows faster than conventional piece dyeing for polyester workwear shades.

4. Cotton's tonnage stays constant for a decade while total fibre output grows. What happens to cotton's market share, and what is this called in trend analysis?
5. State two concrete effects of the global MMF shift on Bangladesh's textile industry, one as an opportunity and one as a risk.

Answer key

1. Doubling time = 0.693 ÷ 0.0583 ≈ **12 years** — twice the growth rate roughly halves the doubling time versus the 3 % industry average (≈ 23 years), which is why functional segments transform product ranges within a decade.
2. Food-grade bottle production itself needs recycled PET (bottle-to-bottle closed loops under packaging rules), and the total bottle stream is finite. As textile brands and bottle makers bid for the same clean flake, price rises and availability tightens; the long-term escape is textile-to-textile chemical recycling, which is still scaling.
3. (i) Water and energy: pigmenting the melt skips the entire aqueous dyeing, washing and drying chain for those shades. (ii) Fastness and uniformity: pigment distributed through the filament cannot wash or rub off unevenly, and lot-to-lot shade variation disappears — decisive for workwear contracts.
4. Its share falls steadily (constant numerator, growing denominator) — a *relative decline despite absolute stability*; exactly cotton's position in Figure 1.
5. Opportunity: expanding into MMF apparel and local filament/staple production opens the larger, faster-growing ≈ 70 % of world demand and shortens lead times versus imported fabric. Risk: continued dependence on imported synthetics exposes the industry to Asian mega-plant pricing and supply shocks, while tightening buyer sustainability rules can exclude non-compliant factories from orders altogether.
